INCH-POUND

MIL-R-83536/16A
AMENDMENT 4
13 May 1999
SUPERSEDING
AMENDMENT 3
30 September 1996

MILITARY SPECIFICATION

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 10 AMPERES, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED, ALL WELDED, TRANSIENT SUPPRESSED DC COIL

This amendment forms a part of MIL-R-83536/16A, dated 13 May 1994, and is approved for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages as listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	11
12	12

PAGE 1

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS" and "MARKING (EITHER SIDE)". Add "SQ" to 1.025 dimension.

PAGE 2

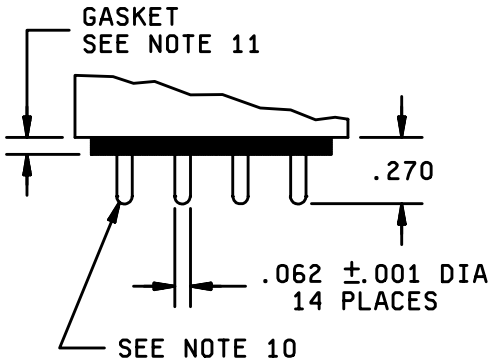
FIGURE 1, side view: Delete "MARKING (EITHER SIDE)".

PAGE 3

FIGURE 1, side view: Delete "MARKING (EITHER SIDE)" and substitute "MARKING (SEE NOTES 8 AND 9)".

PAGE 4

FIGURE 1, socket pin terminal diagram, delete and substitute:



SOCKET PIN TERMINAL

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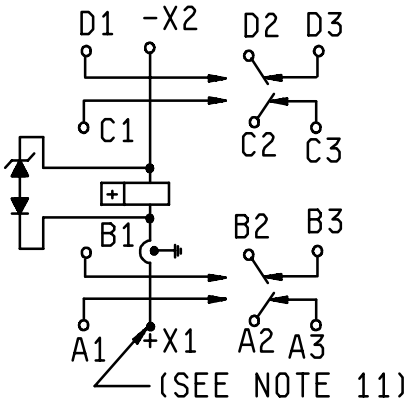
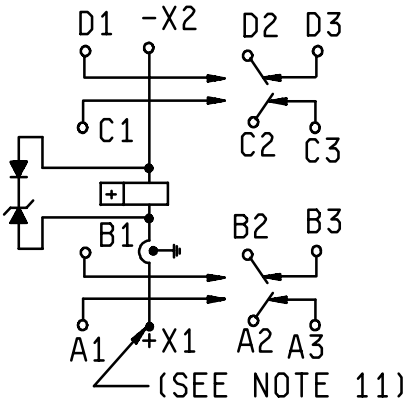
PAGE 8

* FIGURE 2, circuit diagram (track mount), delete and substitute:

"



CIRCUIT DIAGRAM (TRACK MOUNT)



CIRCUIT DIAGRAM (TRACK MOUNT)
(OPTIONAL INTERNAL DIODE CONFIGURATION)

"

PAGE 9

Delete "Contact voltage drop or resistance:" and substitute "Contact voltage drop and resistance:".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA – CC

Review activities:
Air Force - 99

Preparing activity:
DLA - CC

(Project 5945-1043-01)

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TABLE III. Dash numbers and characteristics. 1/

Dash number 2/			Mount	Coil data										
				At 25°C						Over temperature range			Maximum pickup voltage	
Solder pin	Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil resistance (ohms) minimum	Specified pickup voltage (V dc) 4/	Specified hold voltage (V dc) 4/	Specified dropout voltage (V dc) 4/	Specified pickup voltage (V dc) 4/	Specified hold voltage (V dc) 4/	Specified dropout test (V dc) 4/	High temperature test	Continuous current test
				Rated	Max									
001	002	003	No mount	6	7.3	16	3.2	1.6	0.4	4.5	2.3	0.25	5.0	5.7
004	005	006	RVFM	6	7.3	16	3.2	1.6	0.4	4.5	2.3	0.25	5.0	5.7
007	008	---	HFM	6	7.3	16	3.2	1.6	0.4	4.5	2.3	0.25	5.0	5.7
009	010	011	No mount	12	14.5	63	6.5	3.3	0.75	9.0	4.5	0.5	9.9	11.25
012	013	014	RVFM	12	14.5	63	6.5	3.3	0.75	9.0	4.5	0.5	9.9	11.25
015	016	---	HFM	12	14.5	63	6.5	3.3	0.75	9.0	4.5	0.5	9.9	11.25
017	018	019	No mount	28	5/ 29	260	13.5	5.5	2.3	18.0	7.0	1.5	19.8	22.5
020	021	022	RVFM	28	5/ 29	260	13.5	5.5	2.3	18.0	7.0	1.5	19.8	22.5
---	---	6/ 034												
023	024	---	HFM	28	5/ 29	260	13.5	5.5	2.3	18.0	7.0	1.5	19.8	22.5
---	---	025	Track	28	5/ 29	260	13.5	5.5	2.3	18.0	7.0	1.5	19.8	22.5
026	027	028	No mount	48	50	800	24.0	12.0	3.0	36.0	14.0	2.0	38.0	42.0
029	030	031	RVFM	48	50	800	24.0	12.0	3.0	36.0	14.0	2.0	38.0	42.0
032	033	---	HFM	48	50	800	24.0	12.0	3.0	36.0	14.0	2.0	38.0	42.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup, hold, and dropout voltages are for test purposes only and are not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.
6/ Dash number -034 is supplied without internal arc barriers. This relay is not rated for ac loads.

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Dielectric withstanding voltage 3/:

	Track mount		All others	
	Coil to case	All other points	Coil to case	All other points
Sea level:				
Initial	1050	1500	1000	1250
After life	1050	1250	1000	1000
Altitude:				
At 80,000 feet	350	350	350	350
At 300,000 feet	---	---	500	500

Maximum leakage current for track mount only: 100 microamperes rms.

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet (80,000 feet for track mount).

Shock (specified pulse): Applicable, MIL-STD-202, method 213, test condition C, 200 g's for 6 ±1 ms, except horizontal flange mount peak g value shall be 100 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal): Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 10 to 3,000 Hz. Horizontal flange mount shall be 20 g's, 57 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202, method 214, test condition IG (0.4 g^2/Hz, 50 to 2000 Hz). For horizontal flange mount, test condition IE (0.2 g^2/Hz) applies. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to track mount only. Applicable to qualification only. Test in accordance with MIL-STD-202, method 214, test condition IG, 1.5 hours per axis. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Acceleration: Applicable, except 15 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: 0.17 pound (77 grams), 0.2 pound (90.6 grams) for track mount.

Terminal strength:

Solder hook terminals:

Pull force: 10 pounds ±1.0 pound.

3/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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